

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

Name: \_\_\_\_\_ Roll No: \_\_\_\_\_

Date of Conduct: Submission Date: \_\_\_\_\_ Signature of the Lab Tutor:

\_\_\_\_\_

| LAB PERFORMANCE INDICATOR | SUBJECT KNOWLEDGE | DATA ANALYSIS AND INTERPRETATION | ABILITY TO CONDUCT EXPERIMENT |
|---------------------------|-------------------|----------------------------------|-------------------------------|
| SCORE                     |                   |                                  |                               |

**Lab # 4: Data Visualization and Text Analytics using Matplotlib and Word Cloud**

**Duration: 3 Hours**

**Objective:**

The objective of this lab session is to introduce students to data visualization and basic text analytics using Matplotlib and Word Cloud.

Expected Outcomes: By the end of this lab session, students will be able to:

1. Matplotlib: Create and customize different types of plots, including line plots, bar charts, histograms, and scatter plots, to visualize and interpret data.
2. Word Cloud: Process text data, identify frequently occurring words, and create Word Cloud visualizations to explore patterns and major themes in text datasets.
3. Data Interpretation: Analyze visualizations and extract meaningful insights and patterns from text-based data.

**Introduction to Matplotlib**

Matplotlib is a Python library used to create **graphs and visualizations** from data.

In Big Data Analytics, visualization helps us:

- Identify trends
- Detect unusual values/outliers
- Compare categories
- Understand distributions
- Discover relationships between variables

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

- Communicate analytical results

**Basic import:**

- `import matplotlib.pyplot as plt`

**Matplotlib** = the overall visualization library

**pyplot** = a part/module of Matplotlib that provides the functions for making and controlling plots.

**Basic Visualization**

- 1. Line Plot:** A line plot is useful for showing trends over time.
- 2. Bar Chart:** Bar charts are useful for comparing different categories.
- 3. Histogram:** A histogram shows the distribution of numerical data.
- 4. Scatter Plot:** A scatter plot is useful for examining the relationship between two numerical variables.

**Matplotlib with Pandas**

This is the most important part for your Data science and analytics practical, because you have already learned Pandas in Lab 3.

<https://colab.research.google.com/drive/17F2LRGDfR6OCSrNxlAb642UEVvMthCOA?usp=sharing>

(WORK ON THIS NOTEBOOK)

**Word Cloud in Data Science and Analytics**

**What is Word Cloud?**

A Word Cloud is a visual representation of text data where words are displayed in different sizes.

- Frequently occurring words → appear larger
- Less frequent words → appear smaller
- It helps us quickly identify the most common or important words in a large collection of text.

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

- For example, if a dataset contains many posts about Python:

PYTHON  
DATA PYTHON  
ANALYTICS  
CODE DATA  
PYTHON

- The word **Python** appears larger because it occurs more frequently.

### **Why is Word Cloud useful in Data Science and Analytics?**

Big Data can contain a huge amount of **unstructured text**, such as:

- Reddit posts and comments
- Tweets/social media posts
- Customer reviews
- News articles
- Online discussions
- Survey responses
- Chat messages
- Feedback and complaints

Reading thousands or millions of text records manually is difficult.

A Word Cloud provides a **quick visual summary of frequently used words**.

**Suppose we have thousands of customer reviews:**

"The product is excellent..."

"Excellent quality and good service..."

"Good product but delivery was slow..."

"Excellent service..."

A Word Cloud might show:

**EXCELLENT** → very large

**GOOD** → large

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

**PRODUCT** → medium

**SERVICE** → medium

**DELIVERY** → small

This immediately gives us an idea about the **dominant words in the dataset**.

**TASK:**

**DATA SCIENCE AND ANALYTICS**

**WORD CLOUD DATA**

incident

"Phishing emails targeted university employees"

"Malware was detected on multiple computers"

"Suspicious network traffic was identified"

"Phishing attack compromised user credentials"

"Ransomware encrypted important company files"

"Unauthorized login attempts were detected"

"A malicious email contained a harmful attachment"

"Network traffic showed unusual activity"

"Several users reported suspicious emails"

"Malware infected computers through a downloaded file"

"An attacker attempted to steal login credentials"

"Phishing messages were sent to university students"

"A ransomware attack disrupted business operations"

"Security monitoring detected suspicious network connections"

"Users received fake password reset emails"

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

"Malicious software was detected on the server"

"An employee clicked a phishing link"

"Unauthorized access was detected on a database"

"Attackers attempted to compromise user accounts"

"Suspicious files were discovered on multiple systems"

"Phishing emails requested sensitive information"

"Network monitoring detected unusual traffic patterns"

"A malware infection affected several workstations"

"An attacker used stolen credentials to access the network"

"Ransomware caused temporary loss of access to files"

"Security analysts detected a suspicious login"

"Employees received fraudulent emails containing links"

"Malicious code was found on a company server"

"An unauthorized user accessed sensitive data"

"Phishing attacks targeted employees through email"

"Suspicious activity was detected in the network"

"Attackers attempted to install malware on computers"

"A ransomware infection affected the organization"

"Several accounts were locked after failed login attempts"

"Sensitive information was exposed after unauthorized access"

"Security software detected malicious activity"

"Employees reported receiving suspicious messages"

"An attacker attempted to steal confidential information"

"Malware was identified during a security scan"

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

"Network traffic indicated a possible cyber attack"

"Phishing links directed users to a fake website"

"Unauthorized access attempts targeted the database"

"Ransomware encrypted files on several computers"

"Security monitoring detected unusual user activity"

"Attackers sent malicious attachments through email"

"A compromised account was used to access company systems"

"Cybersecurity analysts investigated a suspected malware attack"

"Sensitive credentials were targeted by phishing emails"

"Suspicious network connections were blocked by security systems"

"An organization detected multiple attempted cyber attacks"

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

**EXERCISE**

**Q1: Line Plot — Server Requests**

A server records the number of requests it receives during different hours of the day.

hours = ["9 AM", "10 AM", "11 AM", "12 PM", "1 PM", "2 PM", "3 PM", "4 PM"]

requests = [450, 520, 610, 750, 680, 820, 900, 760]

**Tasks:**

1. Create a line plot using Matplotlib.
2. Add a suitable title.
3. Label the X-axis as **Time**.
4. Label the Y-axis as **Number of Requests**.
5. Add markers to the data points.
6. Add grid lines.
7. Identify the hour when the server received the highest number of requests.

**Question:**

What trend do you observe in the number of server requests throughout the day?

**Q 2: Bar Chart — Storage Usage by Department**

A university stores data from different departments. The amount of storage used is given below:

departments = ["Cybersecurity", "Computer Science", "Electrical", "Software", "Telecom"]

storage = [850, 620, 450, 780, 530]

The storage values are in **GB**.

**Tasks:**

1. Create a bar chart.
2. Add suitable titles and axis labels.
3. Give the bars a suitable color.
4. Add grid lines.
5. Display the department names clearly.

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

6. Identify the department using the most storage.

**Question:**

Which department consumes the largest amount of storage?

**Q3: Histogram — File Sizes**

A data analyst records the sizes of files uploaded to a cloud storage system.

```
file_sizes = [  
    12, 15, 18, 20, 22, 25, 27, 30, 32, 35,  
    38, 40, 42, 45, 48, 50, 52, 55, 58, 60,  
    25, 30, 35, 40, 45, 50, 55, 60, 65, 70  
]
```

The values are in **MB**.

**Tasks:**

1. Create a histogram.
2. Use 6 bins.
3. Add a suitable title.
4. Label the X-axis as **File Size (MB)**.
5. Label the Y-axis as **Frequency**.
6. Add an edge color to the bars.
7. Identify the file-size range containing the most files.

**Question:**

What does the histogram tell you about the distribution of file sizes

**Q 4: Scatter Plot — Number of Users vs Data Generated**

A cloud platform records the number of active users and the amount of data generated each day.

```
users = [100, 200, 300, 400, 500, 600, 700, 800, 900, 1000]
```

```
data_generated = [5, 11, 15, 22, 28, 31, 38, 45, 50, 57]
```

The data generated is measured in **GB**.

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

**Tasks:**

1. Create a scatter plot.
2. Add a suitable title.
3. Label the X-axis as **Number of Users**.
4. Label the Y-axis as **Data Generated (GB)**.
5. Change the marker style.
6. Add grid lines.
7. Observe the relationship between the two variables.

**Question:**

As the number of users increases, what happens to the amount of data generated?

**Q5: Choose the Correct Visualization**

For each situation below, students must decide which visualization is most appropriate:

**Options:**

Line Plot | Bar Chart | Histogram | Scatter Plot

1. Number of website visitors recorded every hour.
2. Number of attacks detected for different attack types.
3. Distribution of ages in a dataset of 1,000 users.
4. Relationship between CPU usage and processing time.
5. Monthly electricity consumption of a data center.
6. Number of files stored by different departments.

**Task:**

Write the appropriate visualization for each situation and give **one reason** for your choice.

**Q6: Word Frequency Analysis**

Using the same cybersecurity dataset:

**Tasks:**

1. Convert all text to lowercase.
2. Split the text into individual words.

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**DATA SCIENCE AND ANALYTICS**  
**(7<sup>th</sup> SEMESTER, 4<sup>TH</sup> Year) LAB EXPERIMENT # 4**

3. Count the frequency of each word.
4. Display the **10 most frequently occurring words**.
5. Create a bar chart showing the top 10 words.

**Q7: Open-Ended Data Science and Analytics Task**

You are working as a **data analyst in a cybersecurity organization**. You have been given a large collection of cybersecurity incident descriptions.



Perform a basic text analytics process using Python.

Your solution must include:

- Loading the dataset using Pandas
- Exploring the dataset
- Finding the top 10 words
- Creating a Word Cloud
- Creating a bar chart of word frequencies
- Identifying at least **three important observations**

**Final Question:**

Based on your analysis, what are the major cybersecurity issues represented in the dataset?